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Biobased, Degradable, and Conjugated Poly(Azomethine)s

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ABSTRACT: Carotenoids are a class of biobased conjugated molecules that bear resemblance to the substructure of polyacetylene, a well-known conductive but insoluble polymer. Solubility is an important physical attribute for processing materials using different techniques. To impart solubility in polymers, alkyl side chains are often included in the molecular design. While these design strategies are well explored in conjugated systems, they have not been implemented with carotenoids as a building block in polymers. Here, we show a series of carotenoid-based polymers with varying side chain lengths to tune solubility. Using carotenoid and *p*-phenylenediamine-based monomers, degradable and biobased poly(azomethine)s were synthesized via imine polycondensation. Maximum solubilities corresponding to the varying alkyl chain lengths were quantitatively determined by ultraviolet-visible (UV-Vis) absorption spectroscopy. Since carotenoids are biobased with known degradation products, the effect of acidic and artificial sunlight -promoted degradation was systematically investigated using UV-Vis spectroscopy, ¹H nuclear magnetic resonance spectroscopy (NMR), infrared (IR) spectroscopy, gel permeation chromatography (GPC), and high-resolution mass spectroscopy (HRMS). Our polymer system was found to have two modes of on-demand degradation, with acid hydrolysis accelerating the rate of polymer degradation and artificial sunlight generating additional degradation products. This work highlights carotenoid monomers as viable candidates in the design of biobased, degradable, and conjugated polymers.

1. INTRODUCTION

Select biobased compounds are natural candidates to be incorporated into conjugated polymer systems due to their intrinsic aromaticity or conjugation, leading to opportunities to create electronic materials with site-specific degradable linkages. The majority of reported biobased compounds for degradable electronics center on directly using the small molecule or readily available derivatives.^{1–5} Compared to their small molecule counterparts, polymers offer extended conjugation lengths, a variety of molecular architectures, tunable morphologies, and tailored degradation sites. More recently, compounds such as indigo^{6–8}, vanillin^{9–12}, and eumelanin^{13,14}, amongst others^{15–17}, have been incorporated into electron-conducting polymers due to their commercial availability and inherent presence in nature.^{18,19} For example, indigo has been used as an acceptor unit to create donor-acceptor semiconducting polymers^{6,7} and divanillin¹⁰ has been co-polymerized with aromatic diamines to create poly(azomethine)s. Carotenoids, which resemble polyacetylene, are of notable interest due to

Traditionally, degradable conjugated polymers are created through labile linkages of conjugated monomeric units.^{35,36} While a wealth of labile linkages are synthetically possible, common examples reported in literature include imine^{37–40}, disulfide⁴¹, or ester⁴² bonds. Compared to disulfide or ester bonds, imine bonds along the polymer backbone may retain conjugation if appropriately incorporated, offering concurrent degradability and delocalization needed for favorable electronic behavior. Many poly(azomethine)s that feature sufficient electron mobility utilize non-biobased monomers^{39,43–46} or involve hazardous reagents in their synthesis such as tin in Stille couplings^{39,45,47}, making them unfavorable to use from an environmental standpoint. Alternatively, poly(azomethine)s via imine condensations allows the polymerization between dialdehydes and diamines using a catalytic amount of acid and a drying agent to scavenge water generated in-situ.

Herein, we describe the synthesis of biobased, degradable, and conjugated poly(azomethine)s from 2,7-dimethyl-2,4,6-octatrienedial, referred to as C10 dialdehyde, and *p*-phenylenediamine derivatives (**Figure 1**). C10 dialdehyde is a compound obtained from the oxidative degradation of β -carotene (**Figure 1a**).^{27–29} It was selected due to its commercial availability and dialdehyde functionality, and thus could be used directly in the polycondensation reaction without further synthetic modifications. From a degradation standpoint, carotenoids are susceptible to UV light, enzymatic or chemical oxidation damage, offering alternative triggers for on-demand degradation when integrated into

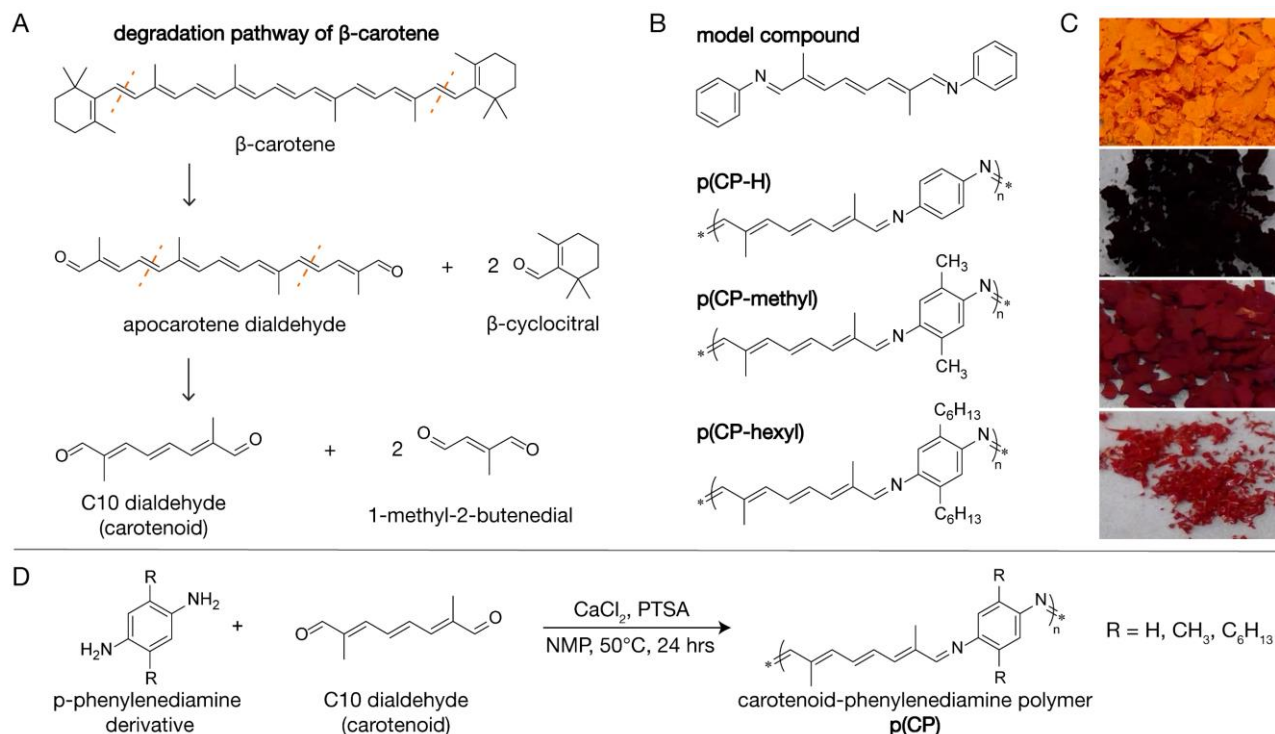


Figure 1. Synthetic overview of biobased, degradable, and conjugated poly(azomethine)s. (a) Oxidative degradation of beta-carotene, which yields C10 dialdehyde, the monomer used in polymerizations. (b) Model compound and three polymers synthesized: **p(CP-H)**, **p(CP-methyl)**, **p(CP-hexyl)**. (c) Photos of the model compound and polymers synthesized. (d) General polymerization scheme for the polycondensation reaction between diamines and dialdehyde to yield the conjugated poly(azomethine)s known as **p(CP)** – polymer carotenoid phenylenediamine.

polymer systems. *p*-Phenylenediamine derivatives were selected because they have been used to synthesize other degradable conjugated polymers^{44,48–50} and because their toxicity^{51–53} has been well documented. Three polymers with different side chains were synthesized via polycondensation (**Figure 1b, 1c, 1d**). Side chain engineering^{54–56} was employed in the design of these polymers as we expected that no side chains would yield an insoluble polymer, making it difficult to characterize and process. To assess the influence of sidechain modification on the solubility, we characterized the polymers by nuclear magnetic resonance (NMR), gel permeation chromatography (GPC), infrared (IR) spectroscopy, and ultraviolet-visible (UV-Vis) absorption spectroscopy. To quantitatively assess the degradation of our polymer system, we monitored the effects of acid hydrolysis and light-mediated oxidation on our polymer system using UV-Vis, ¹H NMR, IR, GPC, and high-resolution mass spectrometry (HRMS). Our results suggest that our polymer system degrades back to C10 dialdehyde and *p*-phenylenediamine derivatives via acid hydrolysis but continues to degrade beyond original monomers in the presence of light.

2. RESULTS AND DISCUSSION

2.1. Synthesis of degradable model compound and conjugated polymers. Three polymers and a model compound were synthesized using the C10 dialdehyde (**Figure 1b**). The model compound, C10 dialdehyde capped with aniline, was used to optimize synthetic conditions, identify characteristic trends in IR, NMR and UV-Vis spectra, as well as study the

degradation mechanism. Due to the commercial availability of most precursors used in this work, only **p(CP-hexyl)** required a diamine monomer to be synthesized. The *p*-phenylenediamine functionalized with 2 hexyl side chains was synthesized using a 2-step reaction, with conditions analogous to those reported in literature for similar compounds.^{57,58} Detailed synthetic procedures and full characterization for the model compound and the synthesized diamine monomer are described in Figures S1–21.

The polymers described in **Figure 1d** were synthesized via polycondensation using a catalytic amount of *p*-toluenesulfonic acid (PTSA) and an excess amount of calcium chloride (CaCl₂) as a drying agent. N-methyl-2-pyrrolidone (NMP) was found to be the best solvent for these polymerizations as it fully solubilized the starting materials and allowed the growing polymer chain to remain soluble. In previous attempts, the polymer would quickly precipitate out of solution during polymerization when using solvents such as tetrahydrofuran (THF) or toluene. Using a catalytic amount of PTSA was crucial to obtain a chain length beyond oligomers and using an excess of calcium chloride ensured that the by-product of water released during the condensation reaction did not prematurely terminate the polymerization.

In total, three polymers were synthesized (**Figure 1b**): **p(CP-H)** served as the control polymer with no side chains, **p(CP-methyl)** contained two methyl side chains, and **p(CP-hexyl)** contained two hexyl side chains. As expected, **p(CP-H)** was a black, insoluble powder (**Figure 1c**) which was characterized by IR, elemental analysis, and ¹³C solid-state

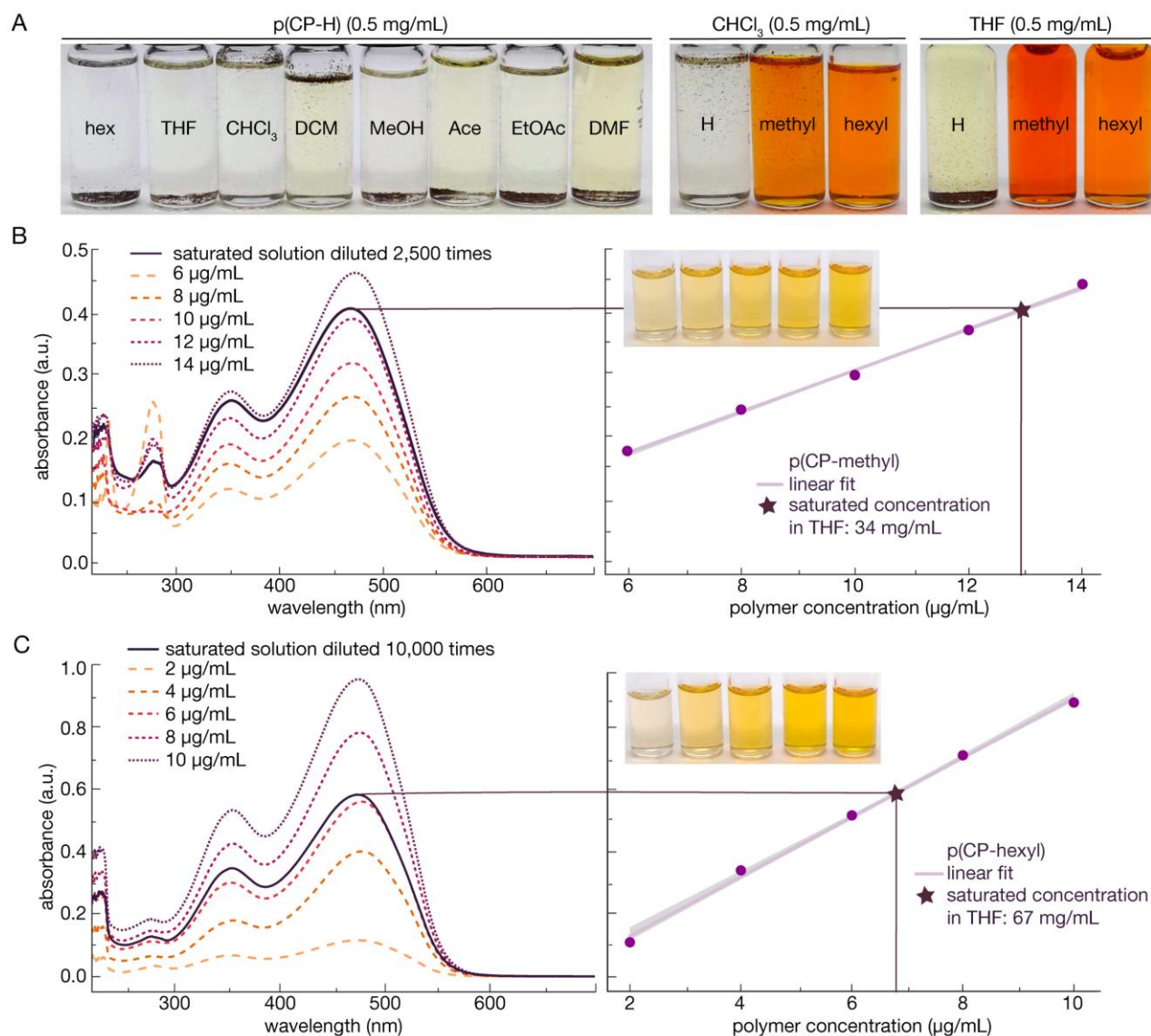


Figure 2. Solution processability of polymers with different side chain lengths. (a) Left: Insolubility of **p(CP-H)** in different solvents including hexanes (hex), tetrahydrofuran (THF), chloroform (CHCl₃), dichloromethane (DCM), methanol (MeOH), acetone (Ace), ethyl acetate (EtOAc) and dimethylformamide (DMF) (0.5 mg/mL) results in a black precipitate. Middle: Solubility comparison between **p(CP-H)**, **p(CP-methyl)**, and **p(CP-hexyl)** in chloroform (0.5 mg/mL). Right: Solubility comparison between **p(CP-H)**, **p(CP-methyl)**, and **p(CP-hexyl)** in THF (0.5 mg/mL). (b) Quantifying the maximal solubility of **p(CP-methyl)**, and (c) **p(CP-hexyl)** in THF. The left panels show the absorption spectra of **p(CP-methyl)**, and **p(CP-hexyl)** at different concentrations (dotted line) and saturated solutions (solid lines) prepared at room temperature, diluted 2,500 and 10,000-fold, respectively. The right panels show a linear fit of the polymer concentrations plotted against their absorption maxima (470 nm), with photos of the solutions prepared. The concentrations of the diluted saturated samples were calculated using the Beer-Lambert law which can be found in the Supporting Information (Fig S34, Tables S1-2).

NMR. In the ¹³C solid-state NMR, **p(CP-H)** exhibits chemical shifts that can be attributed to carbon atoms in the imine, aromatic, alkene, and methyl groups (Figure S23). From IR, C=N and C=O stretches are observed, which is indicative of imine condensation and possible end-group termination with an aldehyde (Figure S24). Adding methyl side chains to create **p(CP-methyl)** improved solubility and led to the formation of a deep red solid (Figure 1c). However, the most significant improvement to solubility came with the addition of hexyl side chains in **p(CP-hexyl)**, which resulted in a

red solid (Figure 1c) that was soluble in most organic solvents.

2.2 Solubility assessments. Synthesizing a soluble polymer is critical for characterization and processability. Therefore, we assessed the solubility of all three polymers in different solvents (Figure 2a). **p(CP-H)** was found to be insoluble in common organic solvents (e.g., CHCl₃, THF, and DMF), whereas both **p(CP-methyl)** and **p(CP-hexyl)** were able to dissolve in solvents such as chloroform and THF, with **p(CP-hexyl)** dissolving better than **p(CP-methyl)** by visual observation. To prove our hypothesis regarding the

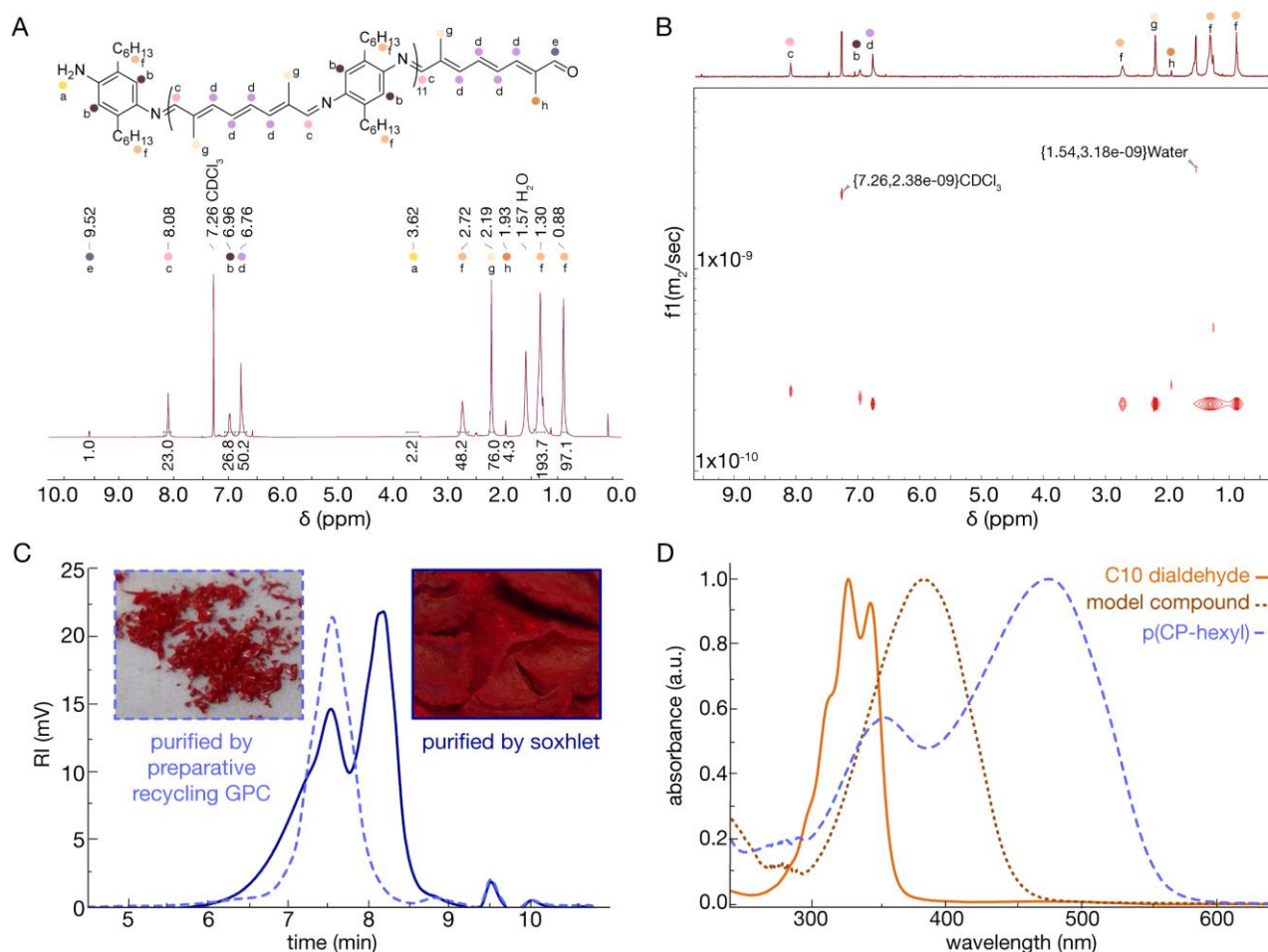


Figure 3. Characterization of **p(CP-hexyl)**. (a) End-group analysis of **p(CP-hexyl)** via ¹H NMR. (b) 2D DOSY map of **p(CP-hexyl)**. Additional data for DOSY can be found in the Supporting Information (Figures S35-39, Tables S3-4). (c) Gel permeation chromatography (GPC) of **p(CP-hexyl)** purified by Soxhlet extraction and purified by preparative recycling GPC. GPC was collected using THF at 40°C. (d) UV-Vis absorbance spectra of C10 dialdehyde, model compound, and **p(CP-hexyl)**. UV-Vis spectra were collected using THF with concentrations of 0.05 mg/mL (see Figure S40 for more details).

greater solubility of p(CP-hexyl) compared to p(CP-methyl), we quantified maximal solubility via UV-Vis absorption spectroscopy according to similar procedures from literature.^{59,60} Linear regressions were created for both p(CP-methyl) and p(CP-hexyl) based on Beer-Lambert's law (Figure 2b, c). A super-saturated sample for each polymer was prepared in THF and its absorbance spectrum taken. From the absorbance value at 470 nm (λ_{max}), and using Beer-Lambert's law, the concentration for each super-saturated solution was determined. For p(CP-methyl), the solubility was found to be 34 ± 1.73 mg/mL, while p(CP-hexyl) was 67 ± 0.29 mg/mL. Based on these values, we carried forward with p(CP-hexyl) for all characterization and degradation studies. Full experimental procedures can be found in Figure S34 and Tables S1-2.

2.3. Molecular Weight Analysis and Optical Characterization. Due to the increased solubility of p(CP-hexyl), it was possible to determine molecular weight by NMR (Figure 3a) and by GPC (Table 1, Figure 3c). Conjugated polymers are often only purified by Soxhlet extraction in which lower molecular weight polymer chains are removed with successive washing of hot solvents. In Figure 3c, the solid GPC trace shows bimodal functionality following Soxhlet purification. The bimodal

feature suggests that p(CP-hexyl) is composed of both high and low molecular weight polymers. To isolate the higher molecular weight polymer chains, preparative recycling GPC in chloroform was used (dotted line in Figure 3c). The higher molecular weight polymer sample was subsequently used for NMR and diffusion-ordered spectroscopy (DOSY) analysis.

End-group analysis was performed using ¹H NMR spectroscopy by setting all integrations with respect to the aldehyde proton (Figure 3a, S36). This was done in assumption that the polymer was terminating with an aldehyde on one end and an amine on the other based on the 1:1 molar ratio used in the polymerization. End-group analysis reveals a degree of polymerization (DP) of 11, which corresponds to a molecular weight (M_w) of 4,800 g/mol. Furthermore, DOSY was used to determine molecular weight (Figure 3b). DOSY gave similar number average molecular weight (M_n) and degree of polymerization (DP) to GPC measurements relative to polystyrene standards with values of 7,700 g/mol and 19, respectively (Table 1). The discrepancy in M_n and DP between the method of end-group analysis and DOSY/GPC results could be from: (1) an over-estimation by GPC/DOSY based on polystyrene calibrants used, (2) an under-estimation via end-group analysis, or (3) a combination

of both. For (1), both DOSY and GPC used polystyrene to build calibration curves from which molecular weight was calculated. Since the structure of our polymer does not bear similarity to polystyrene, it is possible p(CP-hexyl) may interact differently with solvents and GPC columns, resulting in an over-estimation for M_n . For (2), it is possible that not all polymer chains are terminating with an aldehyde proton, and hence the end-group analysis may be under-estimating the true M_w of the polymer sample. Interestingly, p(CP-methyl) gives a DP of 11 by GPC (Table 1), which is analogous to the DP obtained for p(CP-hexyl) by ^1H NMR end-group analysis.

Table 1. GPC Characterization

polymer	M_n (g/mol)	M_w (g/mol)	$\bar{D}$	DP
p(CP-methyl)	2,900	3,700	1.26	11 ^a
p(CP-hexyl)	7,700	10,200	1.31	19 ^a

^a calculated by GPC (solvent: THF, relative to polystyrene standards)

UV-Vis spectroscopy was used to study the effect of conjugation length between the monomer, model compound and polymer (**Figure 3d**) on optoelectronic properties. The C10 dialdehyde monomer has two distinct peak maxima at 328 nm and 344 nm. When increasing conjugation by one imine and phenyl unit on each side of C10 dialdehyde (model compound), we observe a red shift in λ_{max} to 383 nm with a single peak maximum. With p(CP-hexyl), we observe another red shift to 470 nm, with the appearance of a second maximum at 353 nm. The appearance of the second maxima could be due to the presence of alkyl side chains on the diamine unit. Interestingly, the same absorption spectrum is observed for p(CP-methyl) with a slightly lower λ_{max} , suggesting that p(CP-methyl) has fewer repeating units compared to p(CP-hexyl) (Figure S40). This overall trend of increasing λ_{max} with increasing conjugation is consistent with other conjugated systems reported in literature.^{38,61}

2.4. Degradation studies. The degradation of the model compound and polymers were primarily studied using UV-Vis spectroscopy and ^1H NMR. Since the imine bond is cleavable upon acid hydrolysis (**Figure 4a**), hydrochloric acid (HCl) was used as the acid source for our degradation studies. The first study conducted was based on the concentration of acid needed to fully cleave p(CP-hexyl). Degradation onset was imminent upon the addition of 0.25 mM HCl to a solution of 0.013 mg/mL of p(CP-hexyl) in THF, as observed with the presence of a blue shifted λ_{max} , indicating a reduction in conjugation (**Figure 4b**). With successive amounts of acid added, the absorption spectrum changed such that two maxima (328 nm and 344 nm) were visible, showing a return to the C10 dialdehyde monomer upon imine cleavage. This trend was also observed when the same study was conducted with the model compound (Figure S42). Degradation studies using ^1H NMR were also performed to monitor changes in polymer structure with increasing acid concentration (**Figure 4c**). Specifically, we monitored the decrease in integration for the proton indicating the presence of an imine bond and the increase in aldehyde proton integration relative to an internal standard (1,3,5-trimethoxybenzene, TMB). Higher concentrations of acid were required to see the cleavage of the imine bond because this study was

conducted using a higher concentration of p(CP-hexyl) (2 mg/mL in deuterated chloroform (CDCl_3) containing TMB) as opposed to the UV-Vis study. From this study, we observed two aldehyde peaks upon degradation: one corresponding to the C10 dialdehyde monomer, while the other may correspond to oligomers. The results from this study show that a minimum of 270 mM of HCl is required for instantaneous degradation of a 2mg/mL solution of p(CP-hexyl), as indicated by the complete disappearance of the imine bond. To show gradual degradation at a fixed acid concentration over time (6.0 mM HCl), a separate ^1H NMR study was conducted on a 7-day scale (Figures S46-51). The results show the same trend as the study in **Figure 4c**, whereby the imine integration decreased, and aldehyde integration increased over time.

UV light and sunlight have recently been used as a degradation trigger for a diacetylene polymer, which degrades into green upcycling products, offering a mode of degradation that requires less intervention compared to acid hydrolysis.⁶² Therefore, the influence of artificial sunlight on the rate of acid hydrolysis was assessed for p(CP-hexyl). Since C10 dialdehyde is not stable when exposed to light, we wanted to determine how artificial sunlight influences the rate of degradation in conjunction with acid. To do so, four samples were created and subjected to different environmental conditions: one polymer solution kept in the dark, a second polymer solution exposed to continuous artificial sunlight, a third polymer solution kept in the dark with acid, and a fourth polymer solution exposed to continuous artificial sunlight with acid. The relative loss in absorbance at the peak maximum of 470 nm was monitored over a 45-day period (**Figure 4d**). This degradation study showed that the polymer solution kept in the dark was rather stable and showed minimal loss in absorbance at 470 nm. The sample exposed to artificial sunlight showed some tolerance, but still experienced a loss in absorbance over the 45-day period. The samples exposed to acid degraded rapidly, but the sample kept in the dark exhibited a slower degradation rate compared to the sample exposed to artificial sunlight. This degradation rate is illustrated in **Figure 4e**, where the colour for each polymer sample gradually changed upon degradation. By the 34-day mark, the sample exposed to both acid and artificial sunlight had turned cleared, while its counterpart kept in the dark still retained a yellow colour. HRMS of the samples at day 13 of p(CP-hexyl) exposed to UV light with acid revealed that C10 dialdehyde and the diamine monomer were the major degradation products (Table S5). However, since C10 dialdehyde breaks down into epoxides and other dialdehydes when exposed to artificial sunlight (Scheme S2), further investigation into how long it would take for C10 dialdehyde to degrade was done. HRMS taken at day 34 revealed products which are consistent to those reported in literature^{27,28,31} for the oxidative degradation of carotenoids. As a control, a solution of C10 dialdehyde was also exposed to artificial sunlight for a 2-week period, and it too had the same degradation products via HRMS (Table S5). In addition, the absorption spectrum also

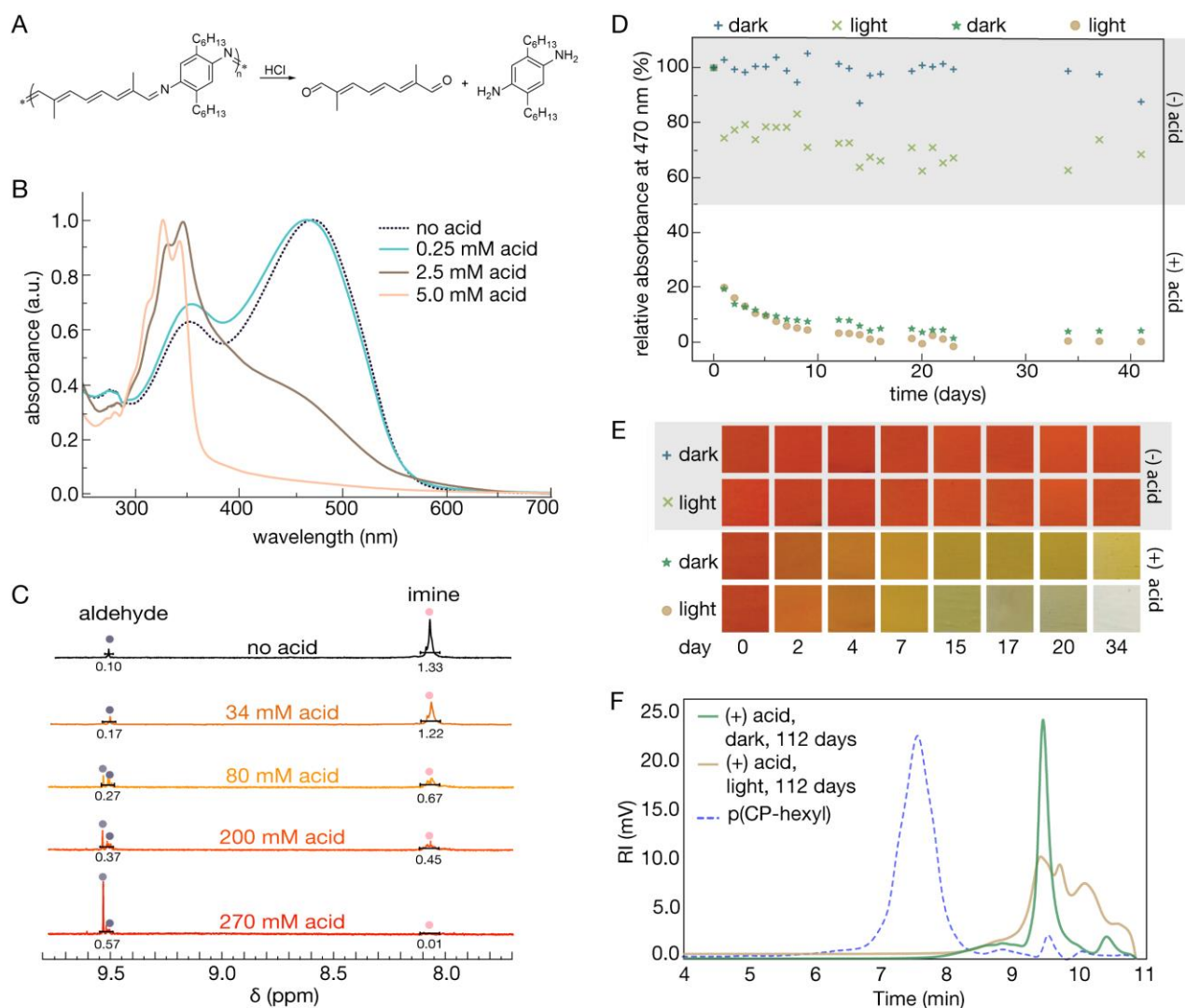


Figure 4. Degradation of **p(CP-hexyl)** monitored by UV-Vis absorption. (a) Proposed mechanism for the acid-hydrolysis of **p(CP-hexyl)**. (b) UV-Vis absorption spectra overlay of **p(CP-hexyl)** in THF treated with various concentrations of HCl (polymer solution of 0.013 mg/mL, with UV-Vis spectrum taken immediately after each addition of HCl). (c) ^1H NMR spectra overlay demonstrating the changes in integration of the proton adjacent to the imine bond and the proton of the aldehyde. Spectra collected using CDCl_3 and TMB as the internal standard. Additional ^1H NMR spectra found in Figures S45-54. (d) Long-term degradation of **p(CP-hexyl)** under the presence of artificial sunlight (polymer solution 0.015 mg/mL) and acid (polymer solution 0.05 mg/mL, 1.2×10^{-7} mM HCl added), with their respective negative controls (solutions kept in the dark). (e) Color change observed upon degradation of each polymer solution over time. (f) GPC traces (solvent: THF) of **p(CP-hexyl)**, the degraded polymer via acid hydrolysis after 112 days kept in the dark, and the degraded polymer via acid hydrolysis after 112 days exposed to artificial sunlight. Additional GPC traces are found in Figures S56-60.

changed dramatically when C10 dialdehyde begins to degrade, and this was the same trend observed in the absorption spectrum of **p(CP-hexyl)** exposed to acid and artificial sunlight (Figure S43-44).

The samples used in **Figures 4d, 4e** were left for an extended period (112 days) to ensure maximal degradation before subsequently analyzing degradation products via GPC, ^1H NMR, and IR. GPC traces of the sample exposed to both acid and artificial sunlight supports the trend observed by UV-Vis and HRMS, in that more degradation products are present when the polymer is exposed to both acid and artificial sunlight compared to the polymer kept in the dark

(**Figure 4f**). The same trend is seen for samples only exposed to artificial sunlight versus kept in the dark without acid (Figure S57). Analysis via ^1H NMR using TMB as an internal standard also demonstrated additional peaks present for the sample exposed to both acid and artificial sunlight compared to the sample exposed to only acid in the dark, suggesting more degradation products are present when the polymer is exposed to light (Figures S52-54). The samples analyzed by IR showed minimal differences between the samples kept in the dark, exposed to artificial sunlight, and acid in the dark (Figure S61-62). However, the sample exposed to both acid and artificial sunlight did have a

significantly different IR spectrum compared to pristine p(CP-hexyl), but was similar to the spectrum of C10 dialdehyde degrading under artificial sunlight after 78 days (Figure S63). By HRMS, the samples (112 days) exposed to artificial sunlight (in acid or without) had the presence of smaller mass fragments compared to samples kept in the dark (Figures S64-S77). All these results corroborate our hypothesis that artificial sunlight does indeed influence the rate of acid hydrolysis and degrades p(CP-hexyl) beyond C10 dialdehyde as supported by ^1H NMR, GPC, IR, and HRMS data. Thus, there is potential to investigate artificial sunlight as an alternative trigger to degrade this polymer.

3. CONCLUSION

In summary, we synthesized one model compound and three conjugated poly(azomethine)s from biobased and commercially available C10 dialdehyde and assessed the significance of alkyl side chains to impart solubility. The improvement in solubility observed when going from methyl to hexyl side chains indicates that side chain engineering is a promising strategy to tune properties to meet application-specific needs such as conductivity, and environmental compatibility. We also systematically monitored the degradation of this polymer system under acidic conditions and studied the impact of artificial sunlight on the rate of degradation. Future work will include improvements to the synthesis of p(CP-hexyl) to yield higher MW polymers, evaluating conductivity, and investigating the recovery of monomers. For degradation studies, mimicking nature using enzymatic degradation pathways would be of interest to explore. Overall, this work lays the foundation for a new class of fully degradable conjugated poly(azomethine)s that uses building blocks from nature.

ASSOCIATED CONTENT

This material is available free of charge via the Internet at <http://pubs.acs.org>. Detailed synthetic procedure, IR, NMR, DOSY, GPC, UV-Vis, and HRMS data (PDF).

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Author Contributions

The manuscript was written through contributions of all authors.

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Notes

The authors declare no competing financial interest.

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ABBREVIATIONS

Ace, acetone; CDCl_3 , deuterated chloroform; CHCl_3 , chloroform; DCM, dichloromethane; DMF, dimethylformamide; DOSY, diffusion-ordered spectroscopy; DP, degree of polymerization; EtOAc, ethyl acetate; GPC, gel permeation chromatography; HCl, hydrochloric acid; Hex, hexanes; HRMS, high-resolution mass spectrometry; IR, infrared; MeOH, methanol; NMP, n-methyl-2-pyrrolidone; NMR, nuclear magnetic resonance; Mn, number average molecular weight; Mw, molecular weight; PTSA, p-toluenesulfonic acid; THF, tetrahydrofuran; TMB, 1,3,5-trimethoxybenzene; UV-Vis, ultraviolet - visible.

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